



Deep Learning Course – Sem 5

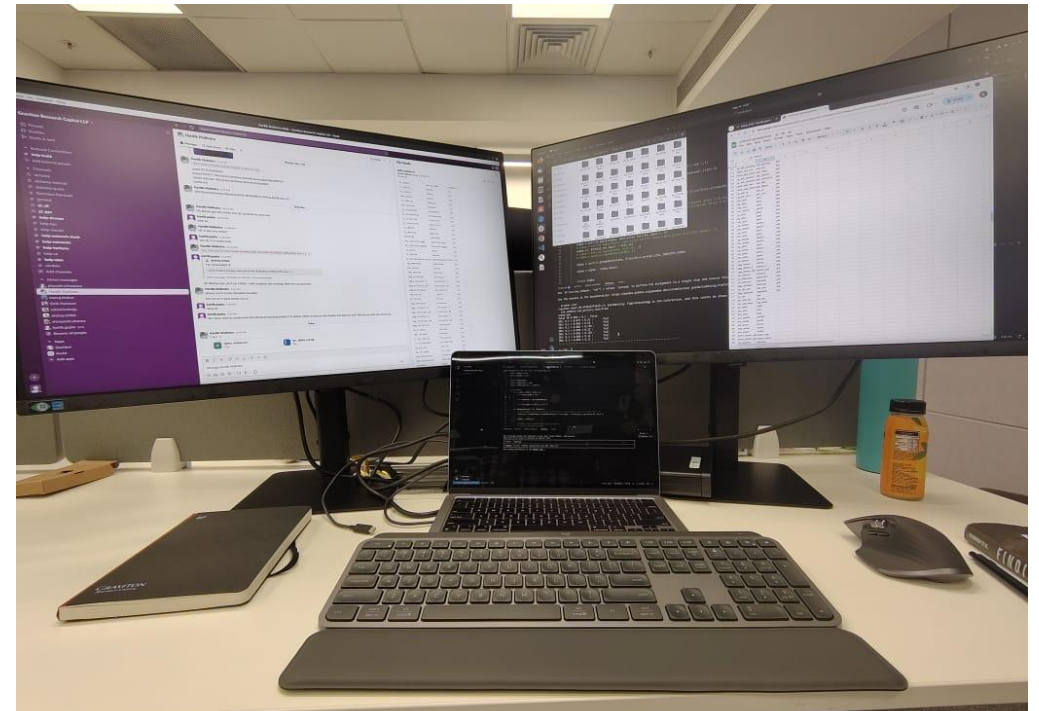
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- **Quantitative Researcher at Graviton Research Capital LLC**, applying statistical and machine learning methods in high-frequency trading.
- Collaborated with IIT Ropar, IIT Bombay and IIT Madras on academic and applied AI initiatives.
- Research interests in **Federated Learning** and **Reinforcement Learning**.



WHERE IT BEGINS

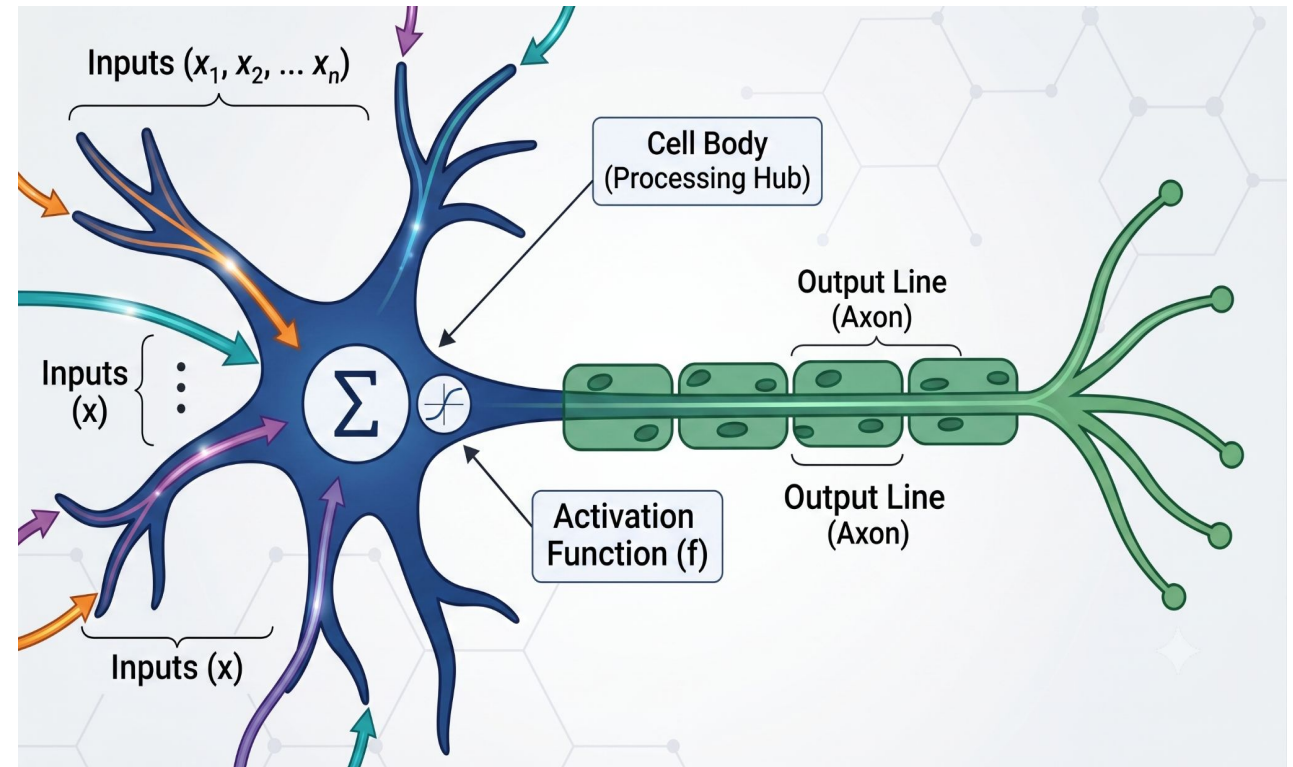
**How can a single brain cell
carry out reasoning?**

THE NEURON

Receive, add up, and fire

A neuron collects signals from many others, **adds them up**, and **fires** only if the total crosses a **threshold**.

- receive
- sum
- fire



A QUESTION

Who first figured this out?

A machine
learning engineer?

A mathematician?

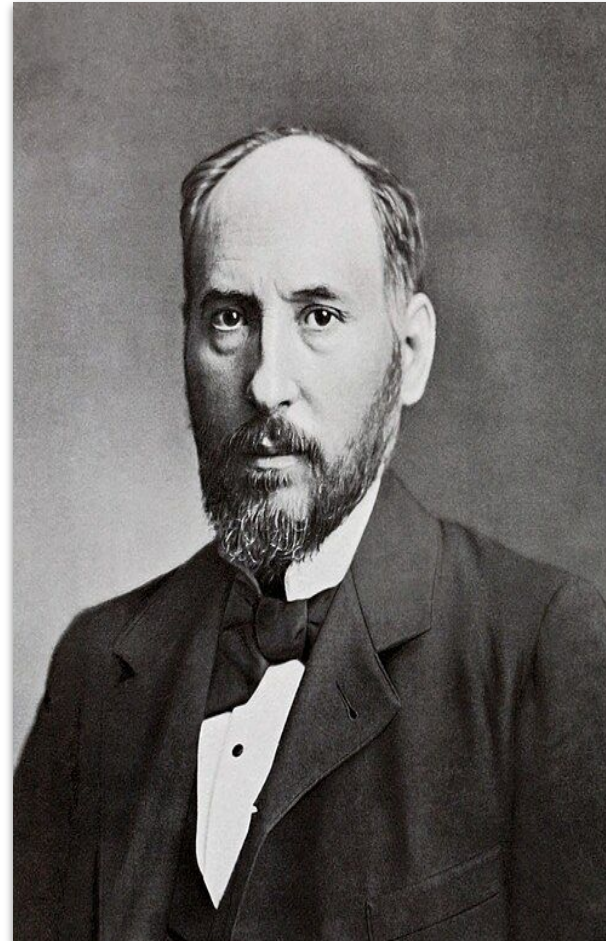
A physicist?

THE ANSWER

A neurologist - Santiago Ramón y Cajal

Around 1888, he showed the brain is built from **separate cells** that pass signals in one direction.

*The father of modern neuroscience
the cell we are about to turn into math.*

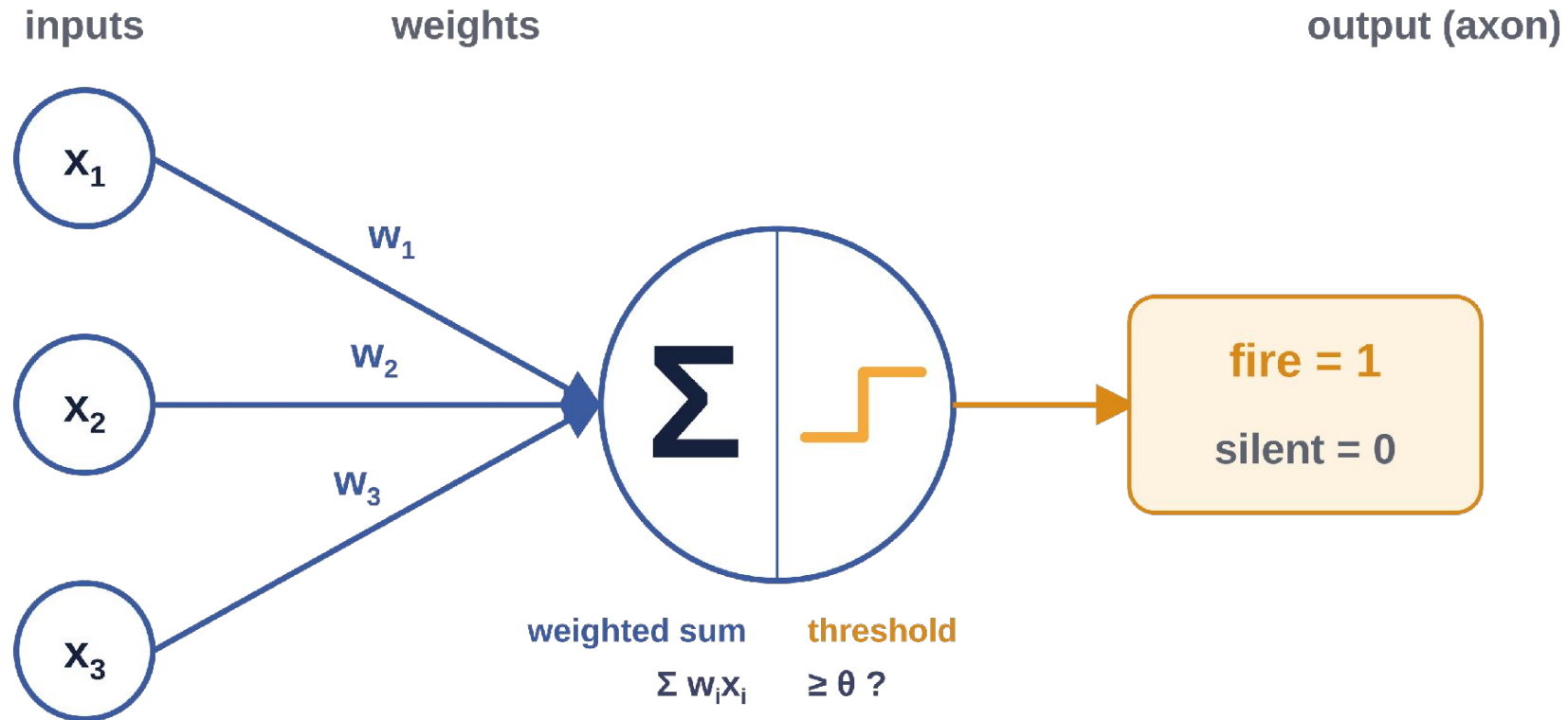


FROM BIOLOGY TO MATH

To carry out logic, there must be math inside the cell. What is it?

THE MATH

A weighted sum, then a threshold



fire if

$$\Sigma w_i x_i \geq \theta$$

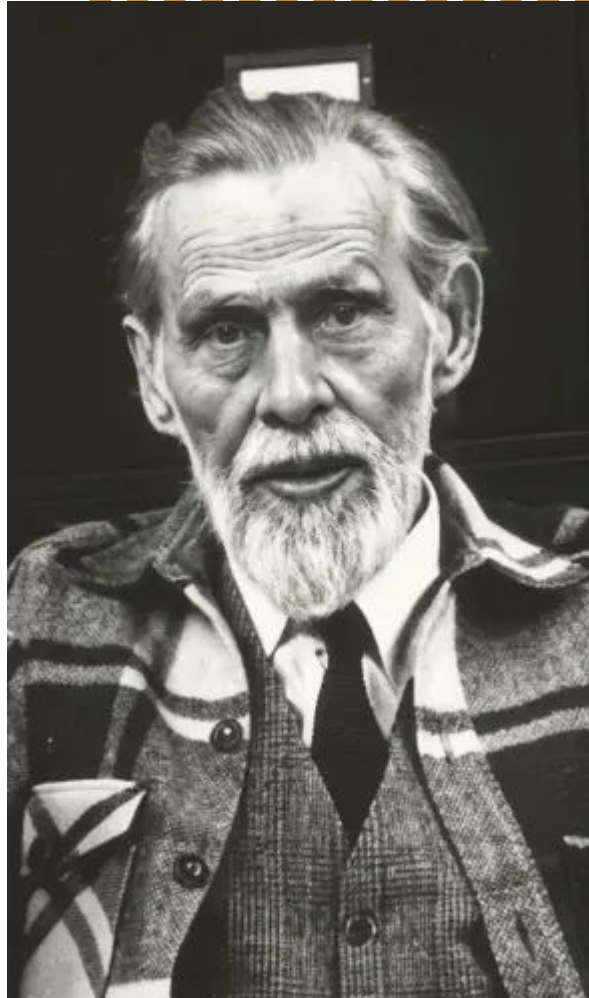
weights scale each input; the **threshold** decides fire or silent.

1943

McCulloch & Pitts write it as a model

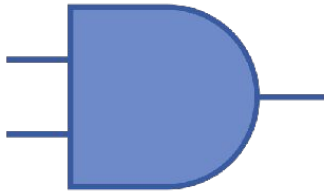
They turned the sum-and-threshold cell into **exact mathematics** and asked one question: **what can it compute?**

*The thing every computer is built from:
logic gates.*



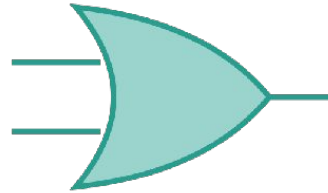
What are logic gates?

AND



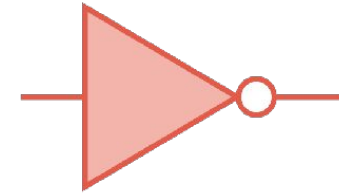
a	b	out
0	0	0
0	1	0
1	0	0
1	1	1

OR



a	b	out
0	0	0
0	1	1
1	0	1
1	1	1

NOT

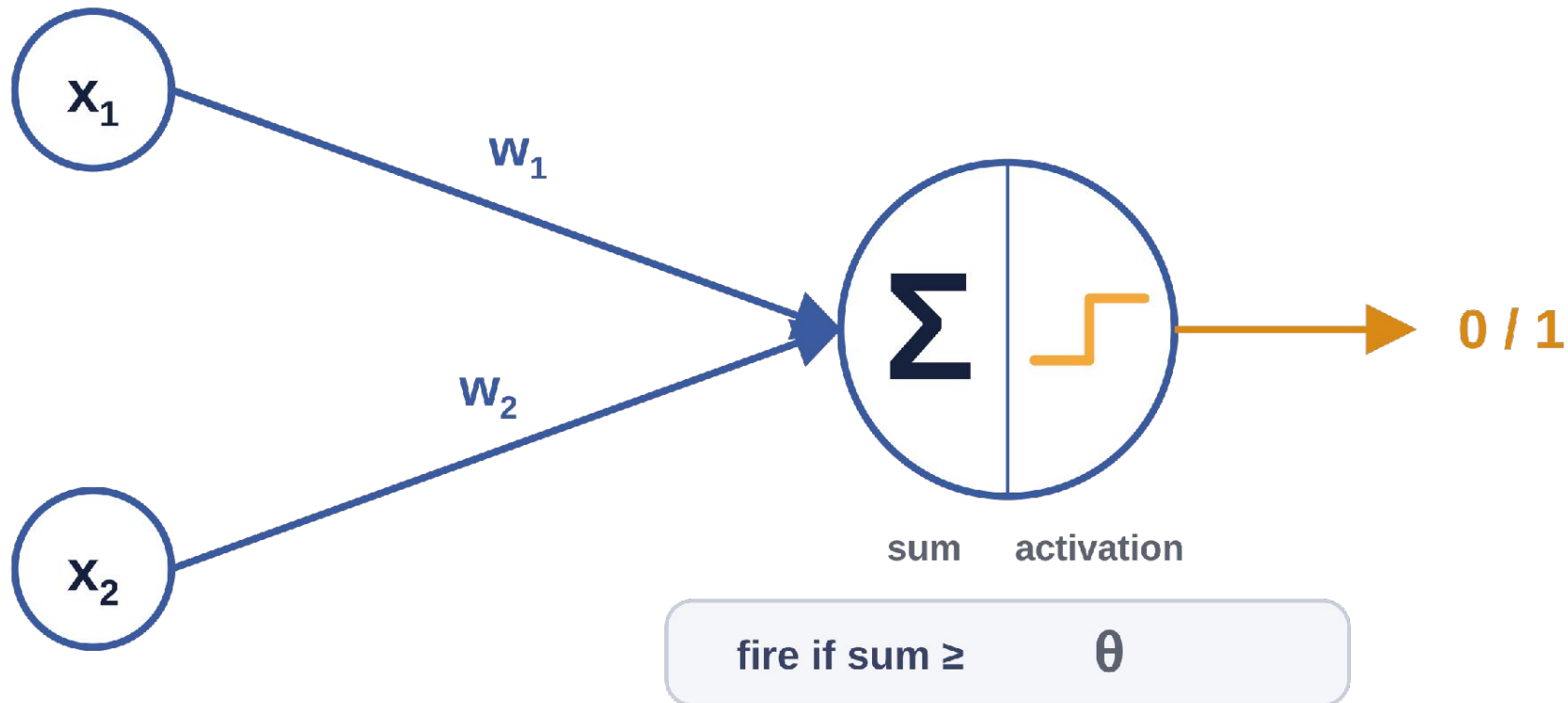


in	out
0	1
1	0

Stack enough of these and you can build any computer — arithmetic, memory, everything.

YOUR TURN

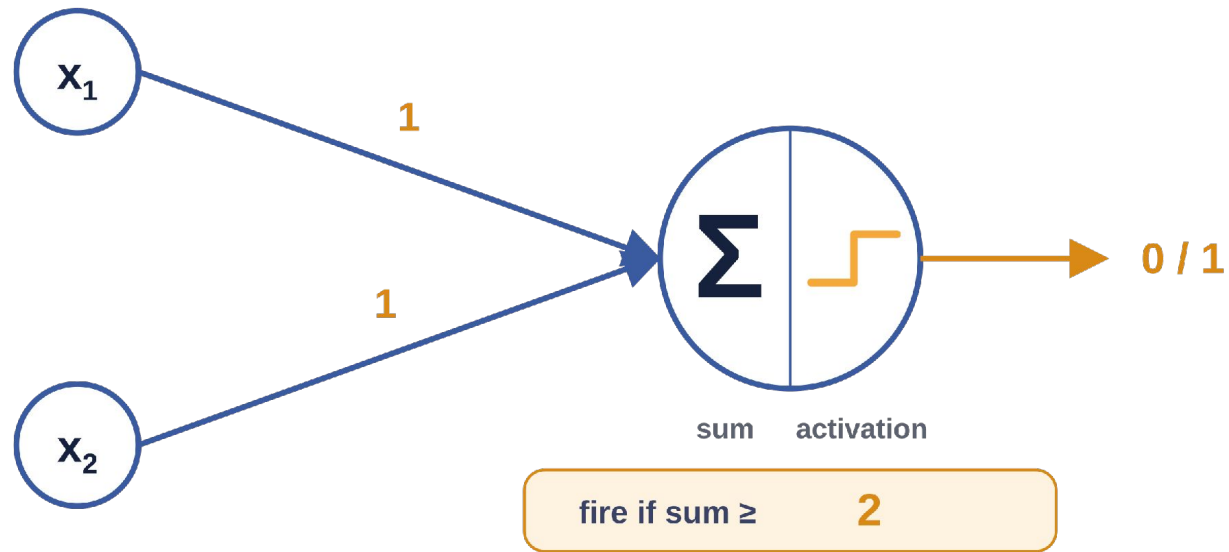
Exercise: make it an AND gate



a	b	out
0	0	0
0	1	0
1	0	0
1	1	1

ANSWER

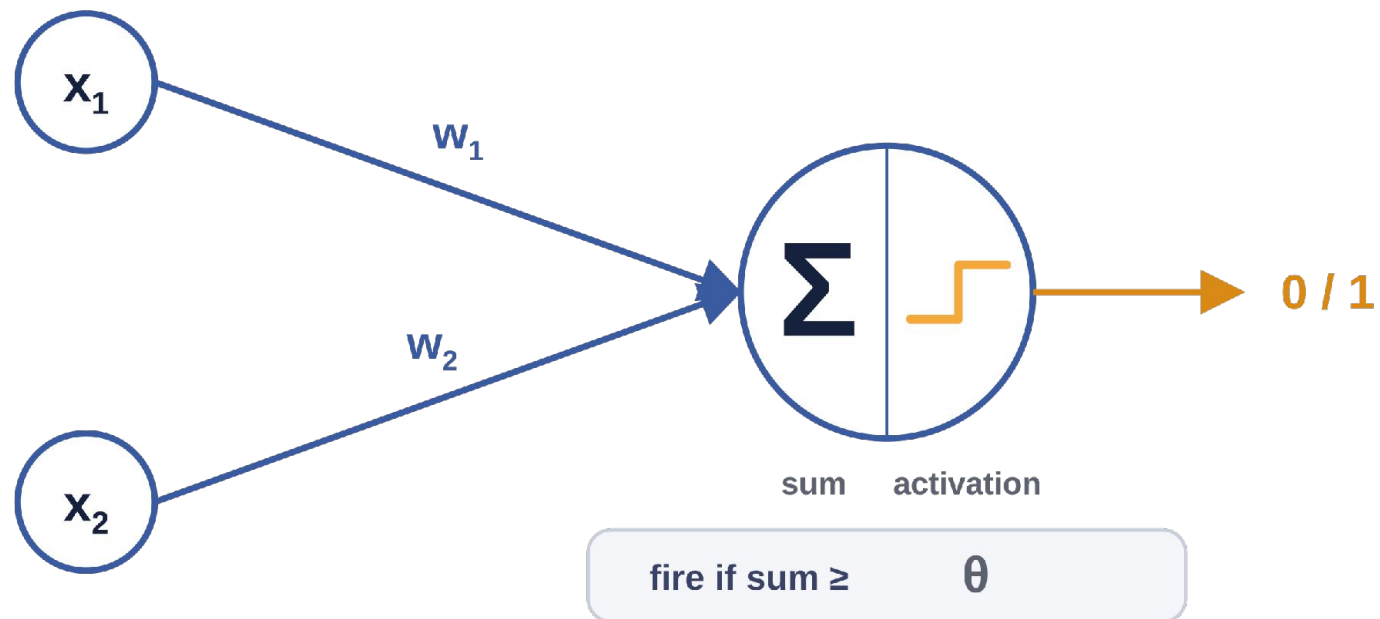
AND: $w_1 = 1$, $w_2 = 1$, threshold ≥ 2



x_1	x_2	sum	out
0	0	0	0
0	1	1	0
1	0	1	0
1	1	2	1

YOUR TURN

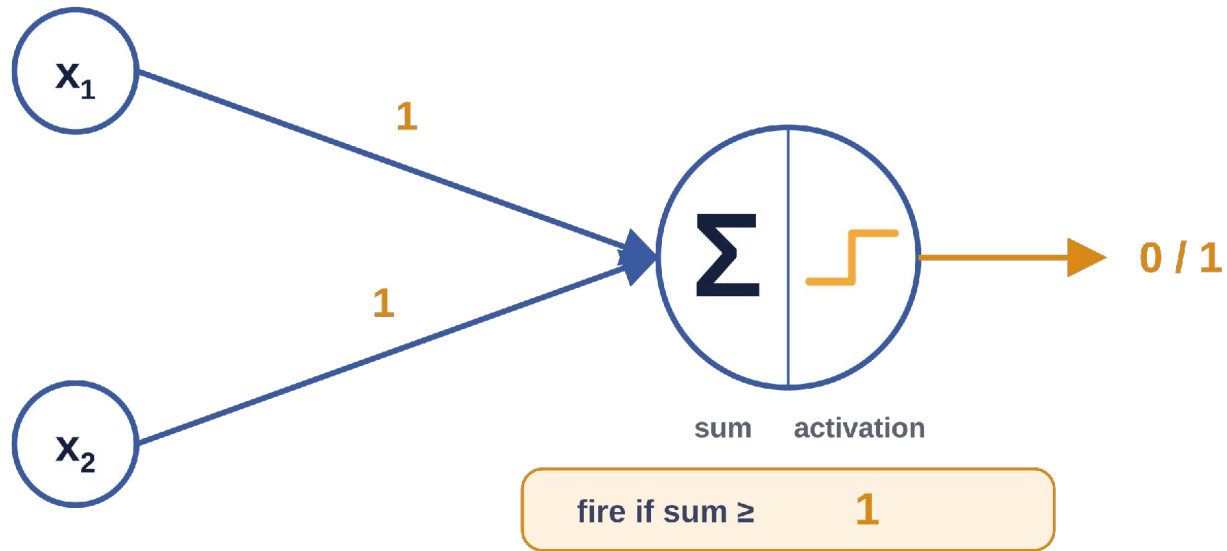
Exercise: now make it an OR gate



a	b	out
0	0	0
0	1	1
1	0	1
1	1	1

ANSWER

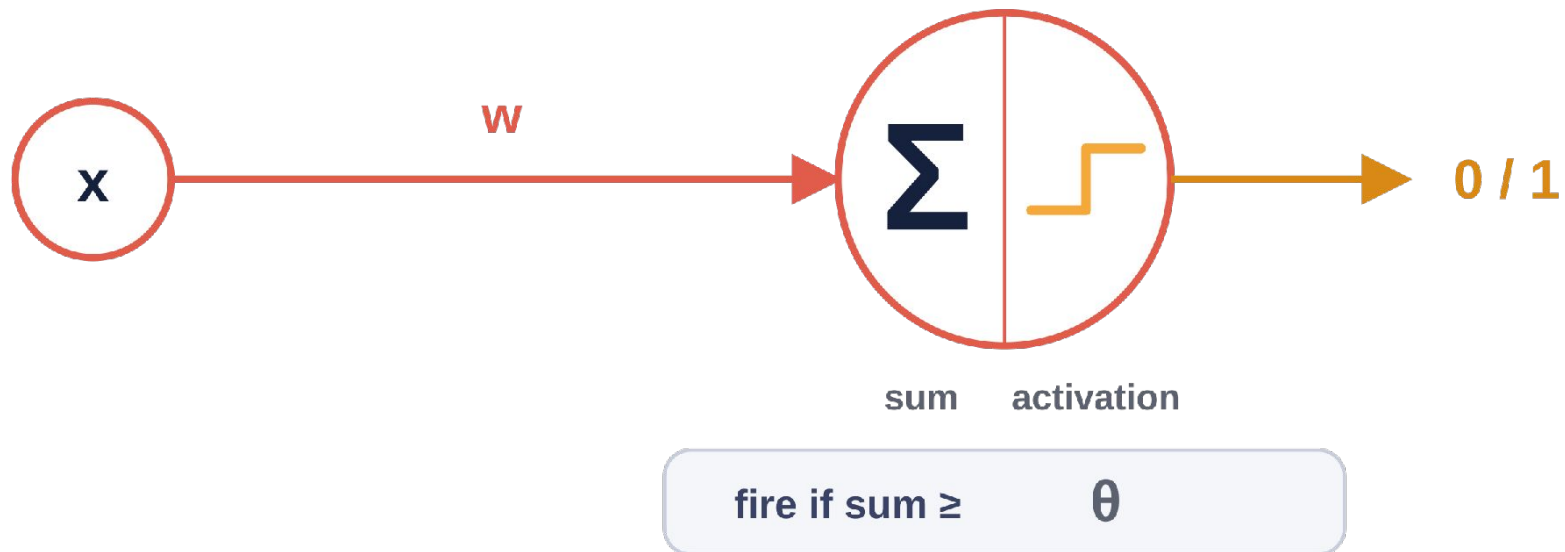
OR: $w_1 = 1$, $w_2 = 1$, threshold ≥ 1



x_1	x_2	sum	out
0	0	0	0
0	1	1	1
1	0	1	1
1	1	2	1

YOUR TURN

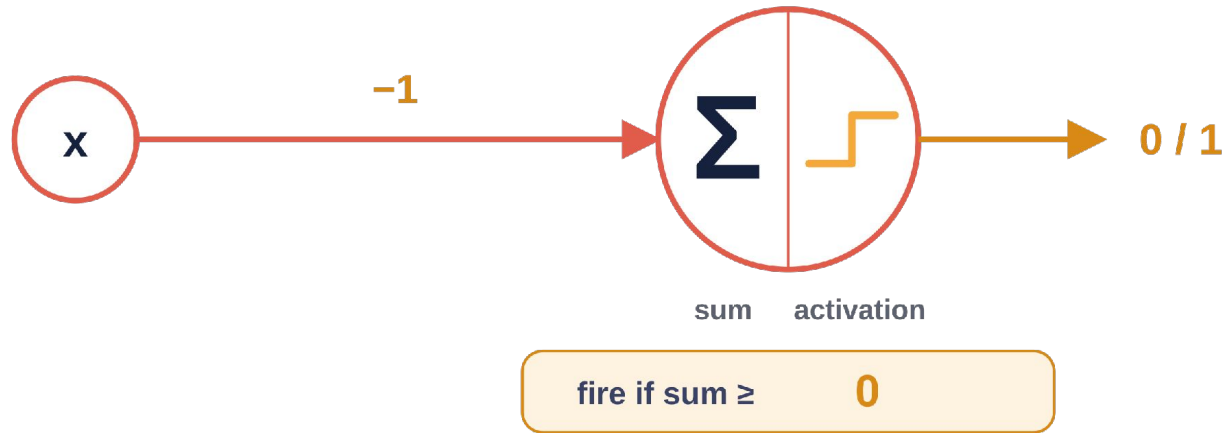
Exercise: one input - make it NOT



in	out
0	1
1	0

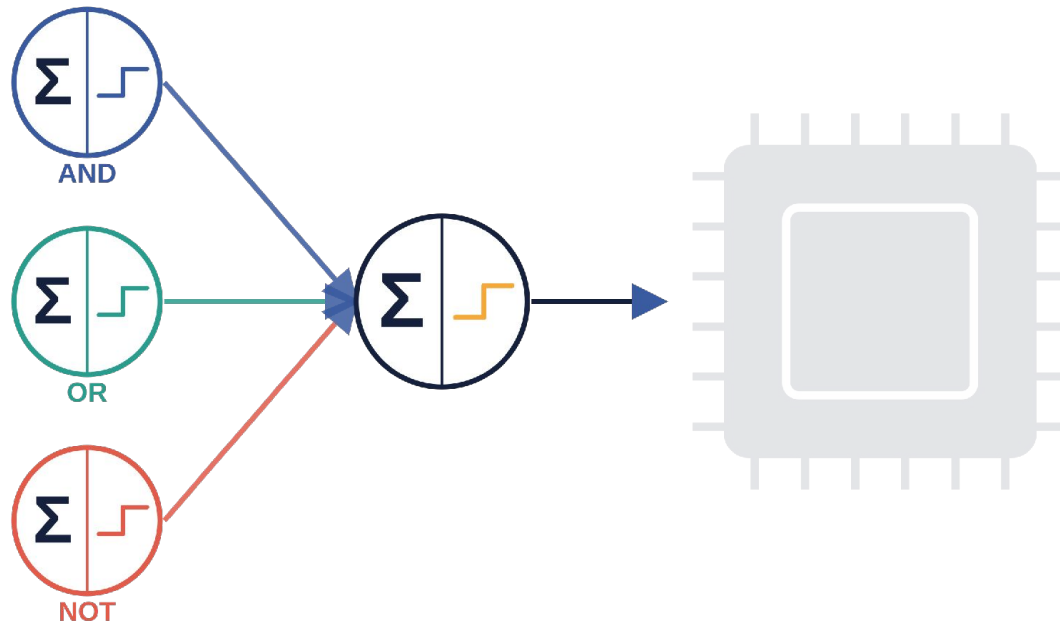
ANSWER

NOT: $w = -1$, threshold ≥ 0



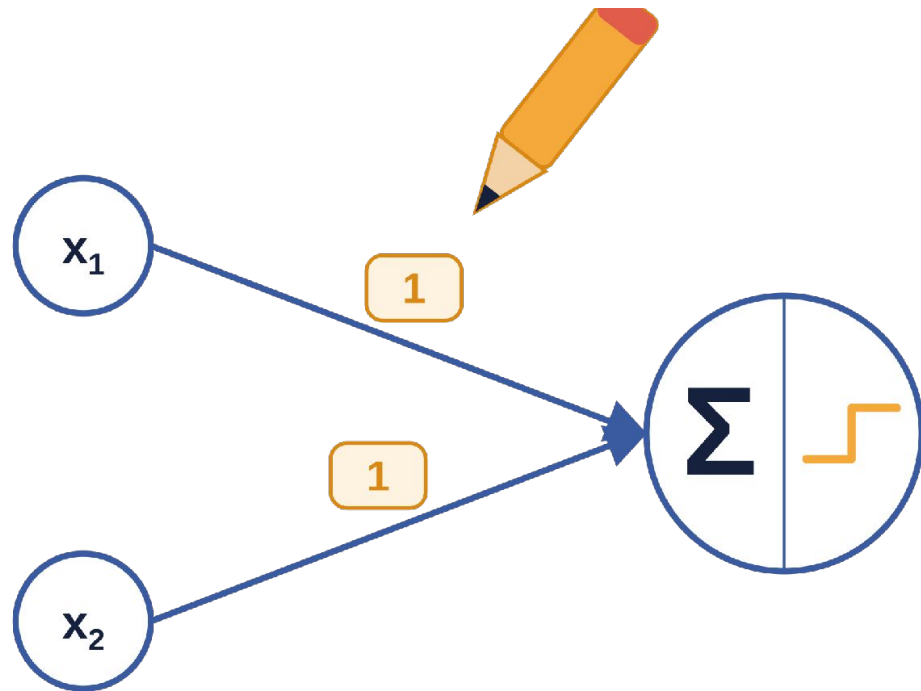
x	sum	out
0	0	1
1	-1	0

One cell can be any gate



AND, OR, NOT in one cell.
So networks of these cells can
build **any logic circuit at all.**

But who chose those weights?



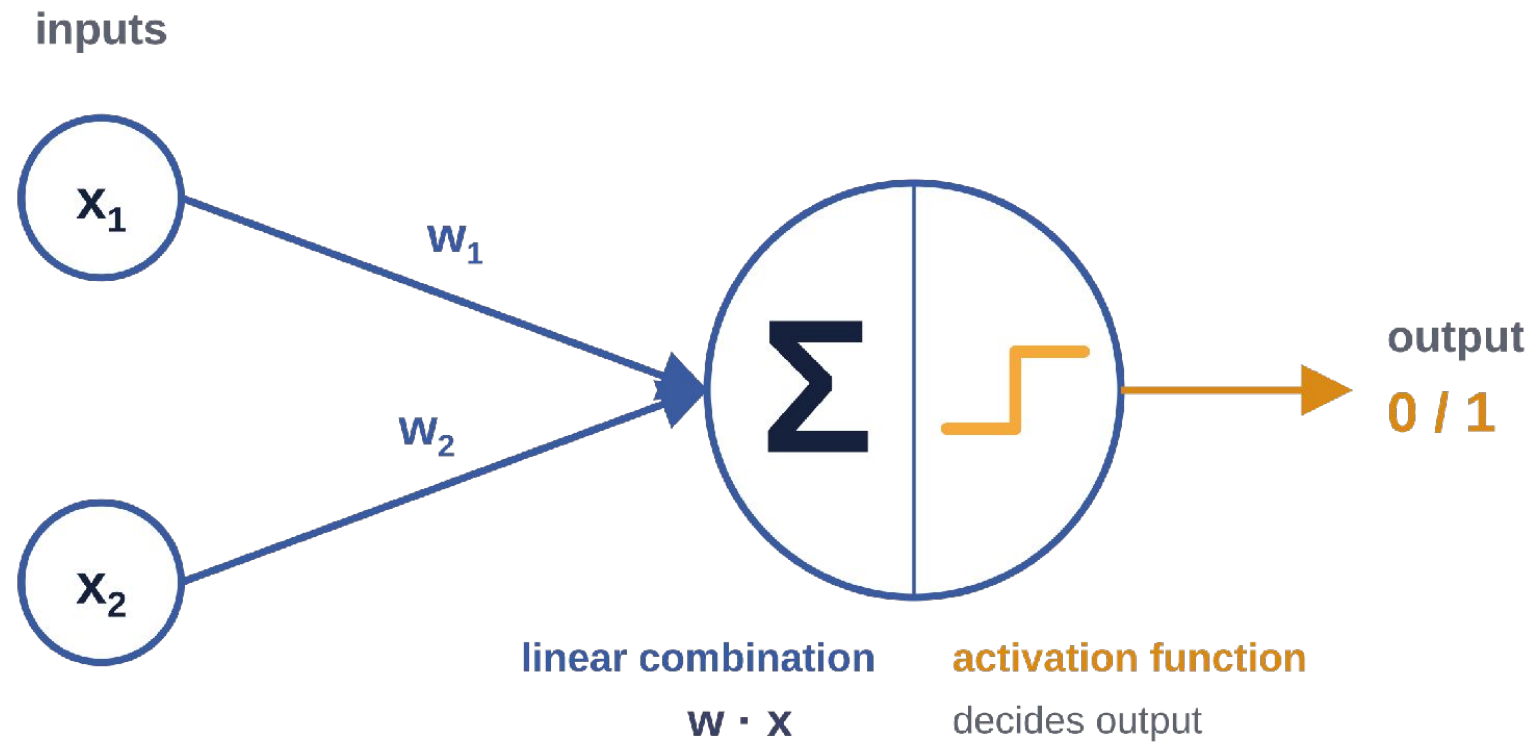
Right now: you did - by hand.

The neuron computes, but nothing changed from experience. It has not **learned** anything.

In a real network, no one sets them.

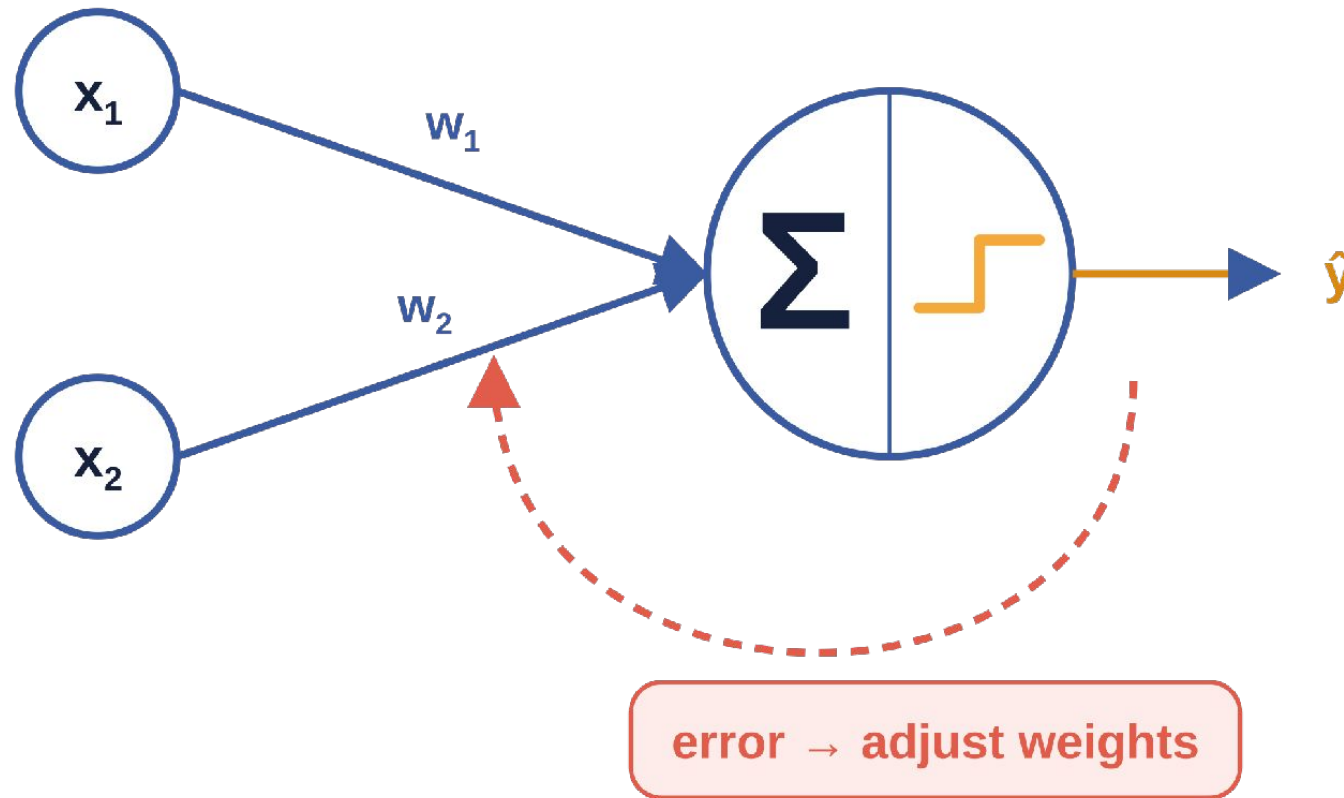
The weights are **dynamic** - chosen automatically, adjusting themselves from data as the model **learns**.

Forward propagation



Linear combination mixes the inputs. **Activation** decides the final output.

Made a mistake? Go back and adjust



This is backward propagation

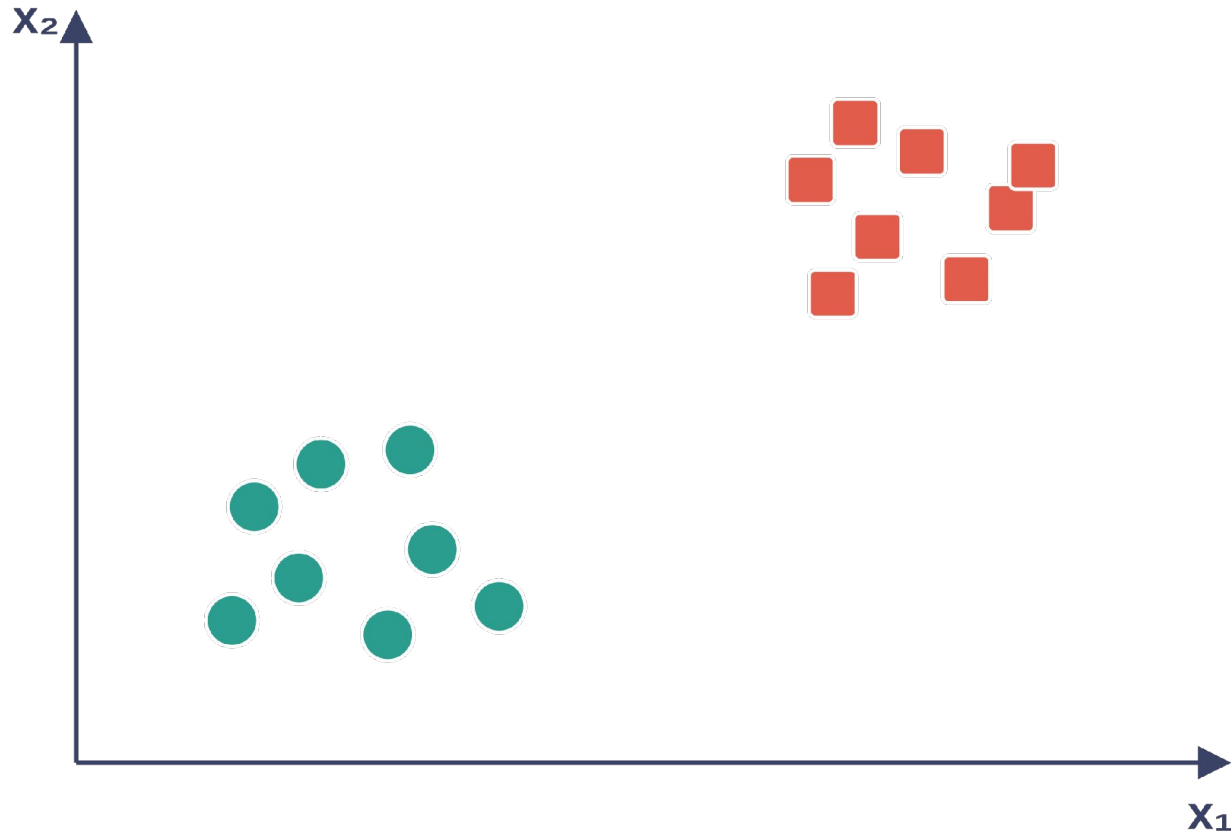
A CHANGE OF DIRECTION

The output is only 0 or 1



Not a number - a category. **That makes the perceptron a classifier.**

Two features, two classes



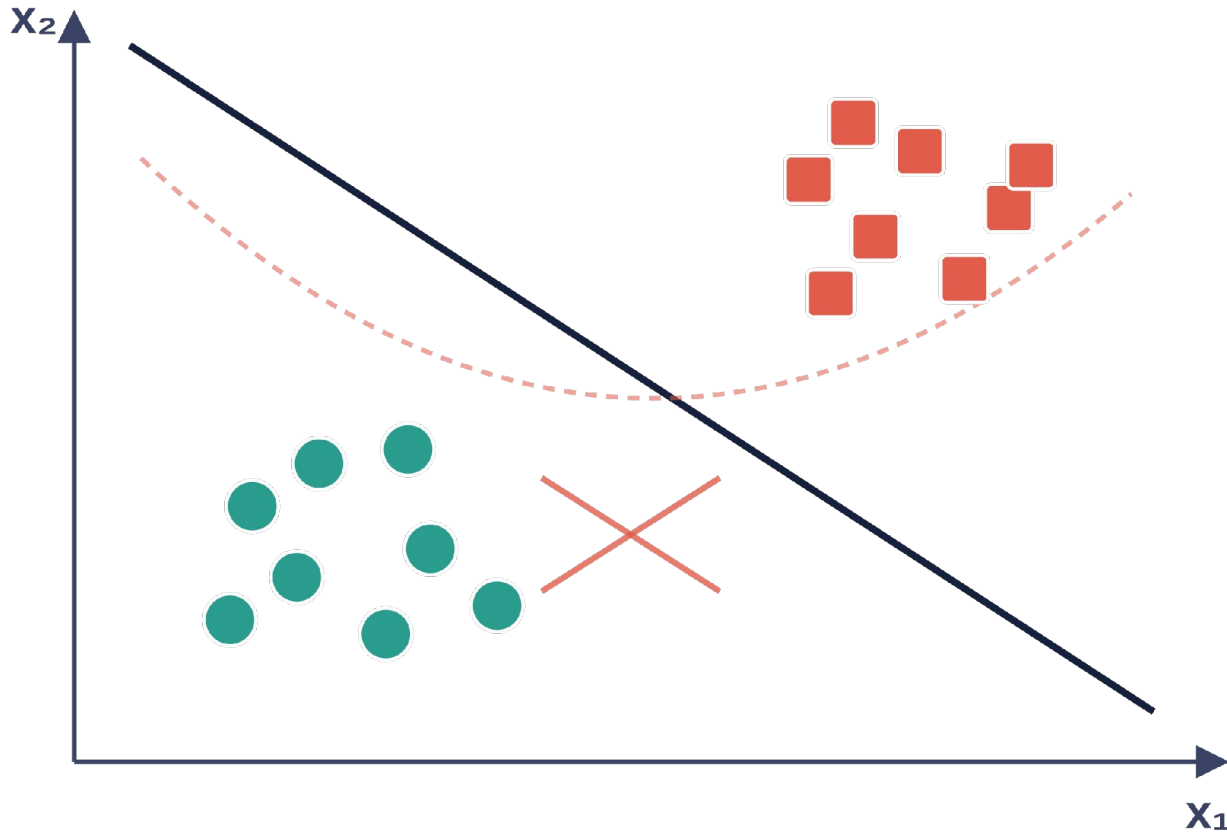
Each point is one example with two inputs (x_1 , x_2).

The perceptron's job: **draw a boundary** that separates the colours.

Get the boundary right, and it answers correctly on every point.

THE CLAIM WE MUST PROVE

A single perceptron draws only a straight line. Why?



A curved one is **not available** to a single perceptron. Why?

RECAP

What the perceptron does so far

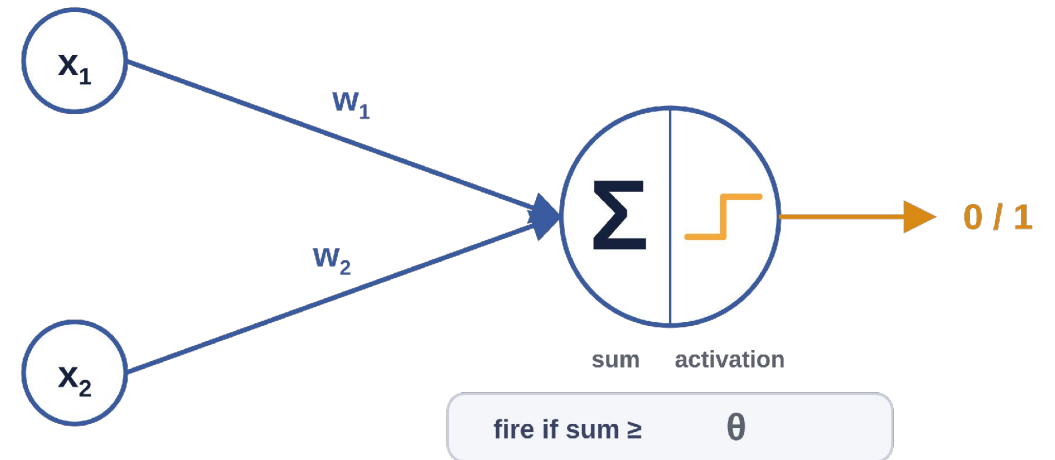
Step 1 — form a linear combination of the inputs:

$$w_1x_1 + w_2x_2$$

Step 2 — compare it with a threshold θ :

output 1 if $w_1x_1 + w_2x_2 \geq \theta$

output 0 otherwise



We will now show what shape this decision makes on the plane.

A SMALL REWRITE

Move the threshold to the other side

$$w_1x_1 + w_2x_2 \geq \theta$$

$$w_1x_1 + w_2x_2 - \theta \geq 0$$

let $b = -\theta$ (b is called the bias)

$$w_1x_1 + w_2x_2 + b \geq 0$$

Nothing has changed — it is the same rule, written more conveniently.

THE DECISION

The whole decision is a sign check

$w_1x_1 + w_2x_2 + b > 0$ → **output 1** (class 1)

$w_1x_1 + w_2x_2 + b = 0$ → **the boundary** (between the two)

$w_1x_1 + w_2x_2 + b < 0$ → **output 0** (class 0)

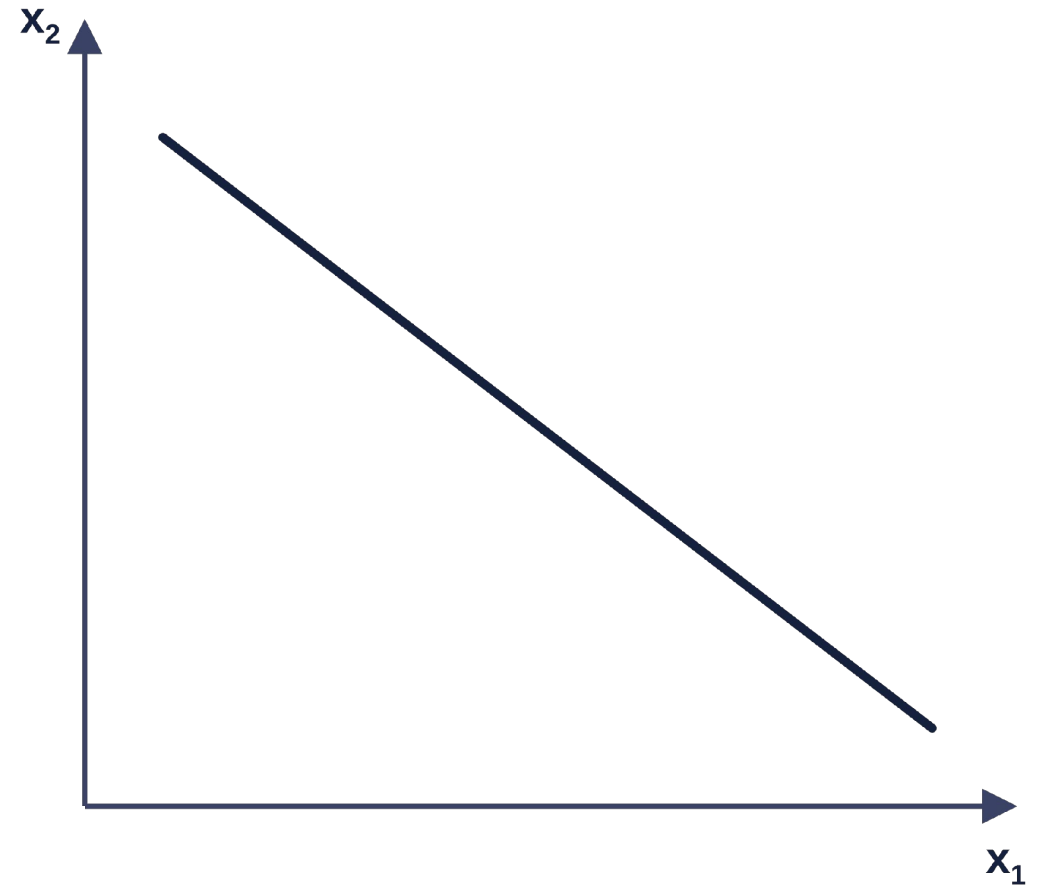
THE BOUNDARY

What is the boundary?

$$w_1x_1 + w_2x_2 + b = 0$$

This is the equation of which geometric figure?

A straight line.



A CONCRETE EXAMPLE

Put in concrete numbers

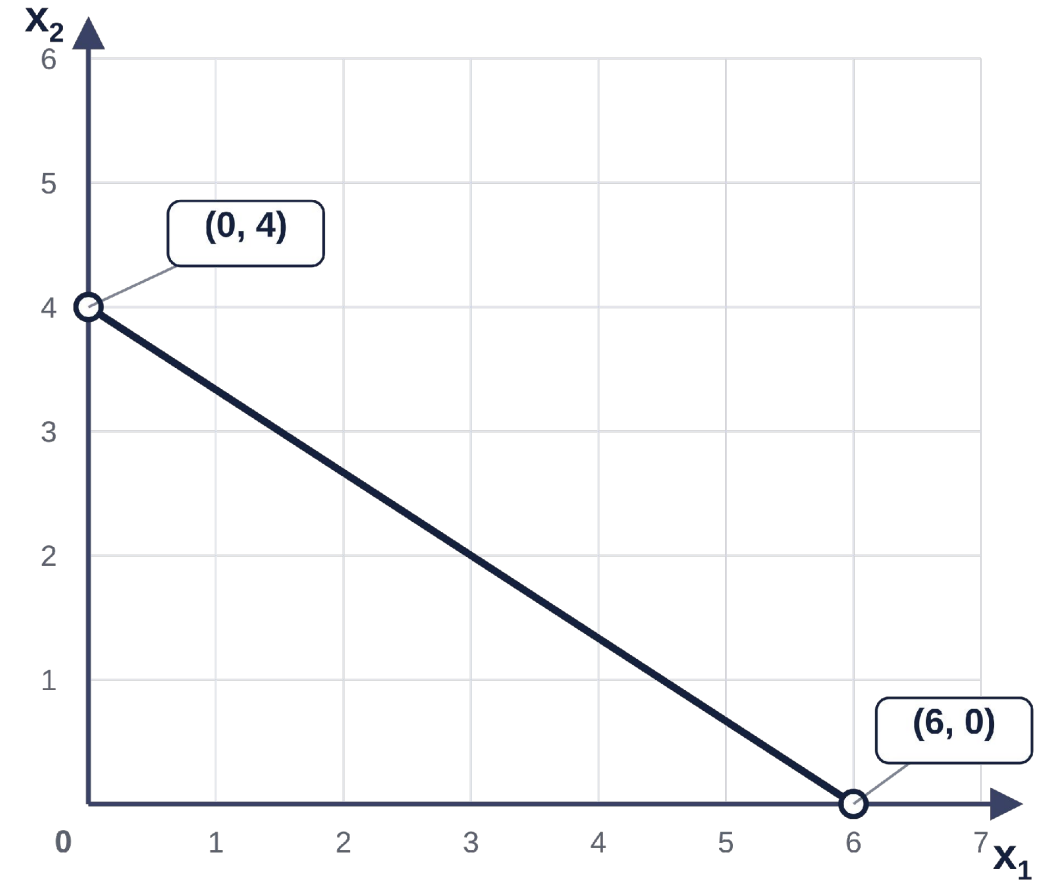
Take $w_1 = 2$, $w_2 = 3$, $b = -12$

Substitute:

$$2x_1 + 3x_2 - 12 \geq 0$$

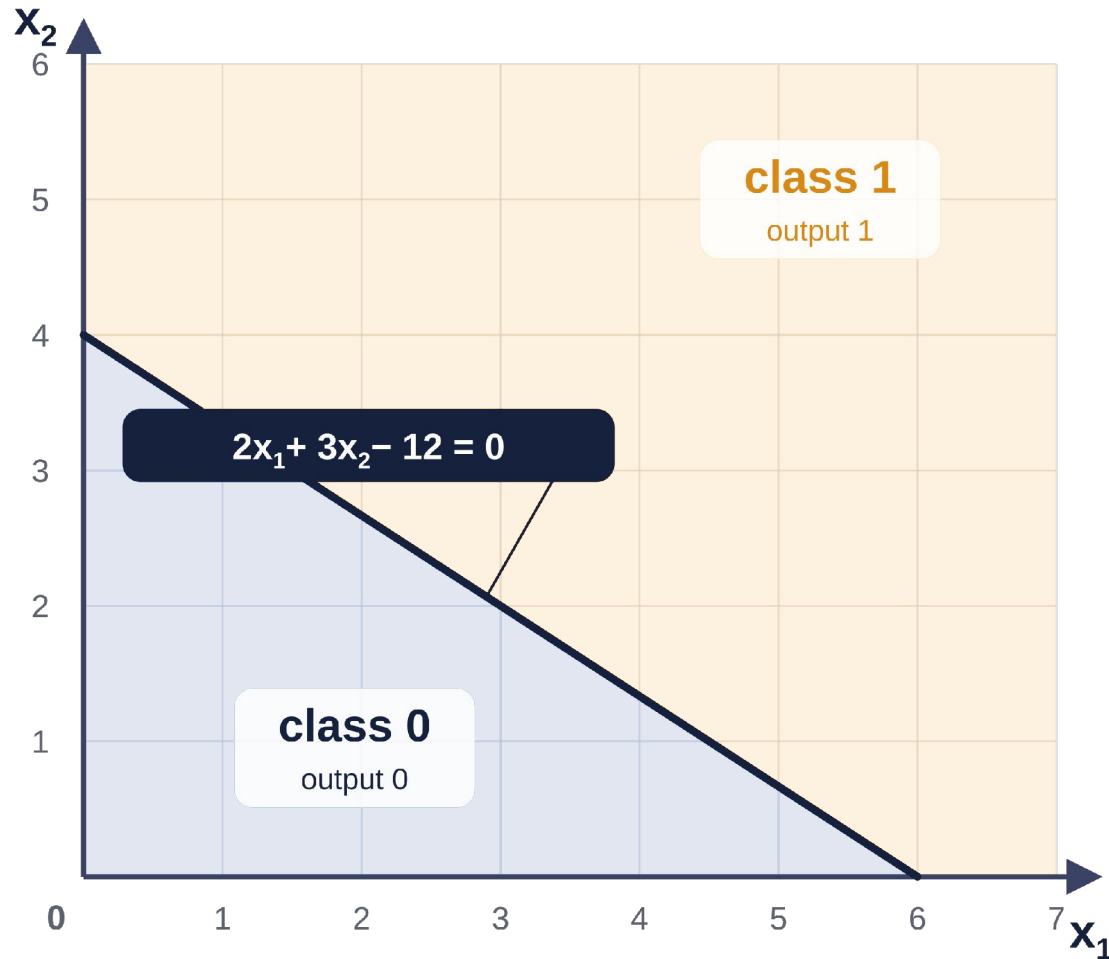
Boundary (set it to 0):

$$2x_1 + 3x_2 - 12 = 0$$



THE FULL PICTURE

The line separates the plane into two classes



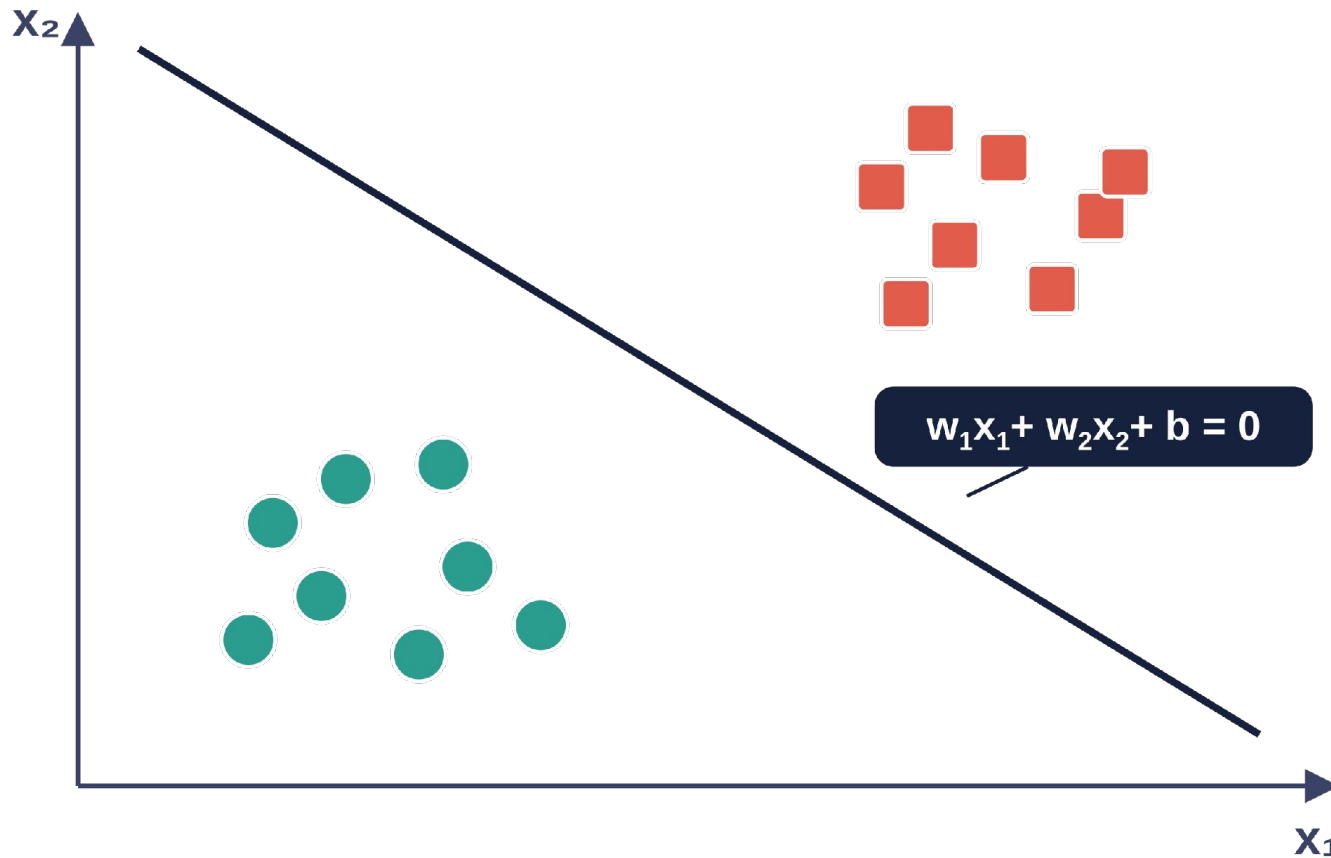
● above $\rightarrow > 0 \rightarrow$ class 1

● on $\rightarrow = 0 \rightarrow$ boundary

● below $\rightarrow < 0 \rightarrow$ class 0

One straight cut, two classes. This is a linear separation.

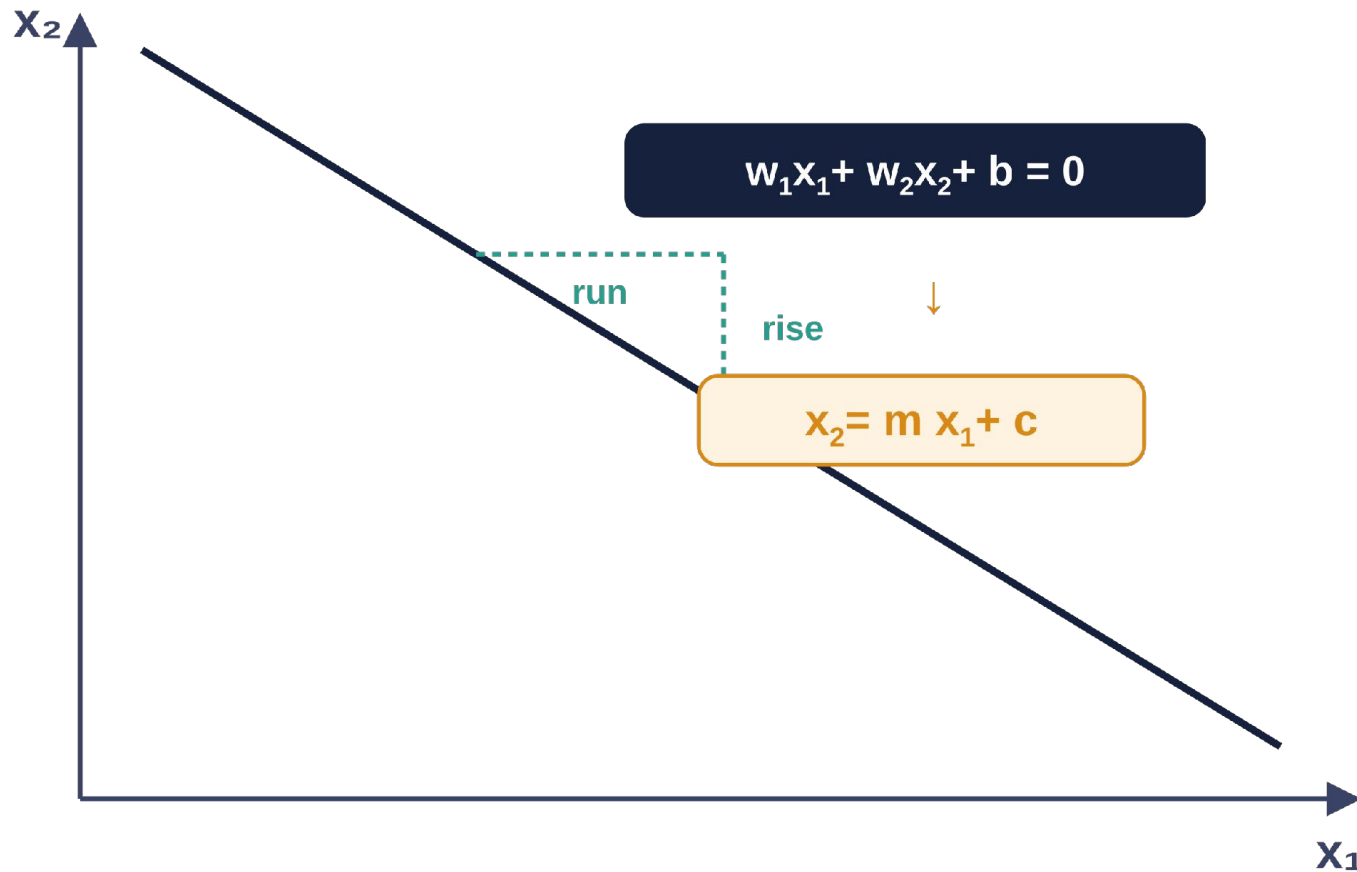
The boundary is where the sum is zero



The boundary is the fence — the exact place the decision flips.

That is precisely where the linear combination equals zero.

That equation is a straight line

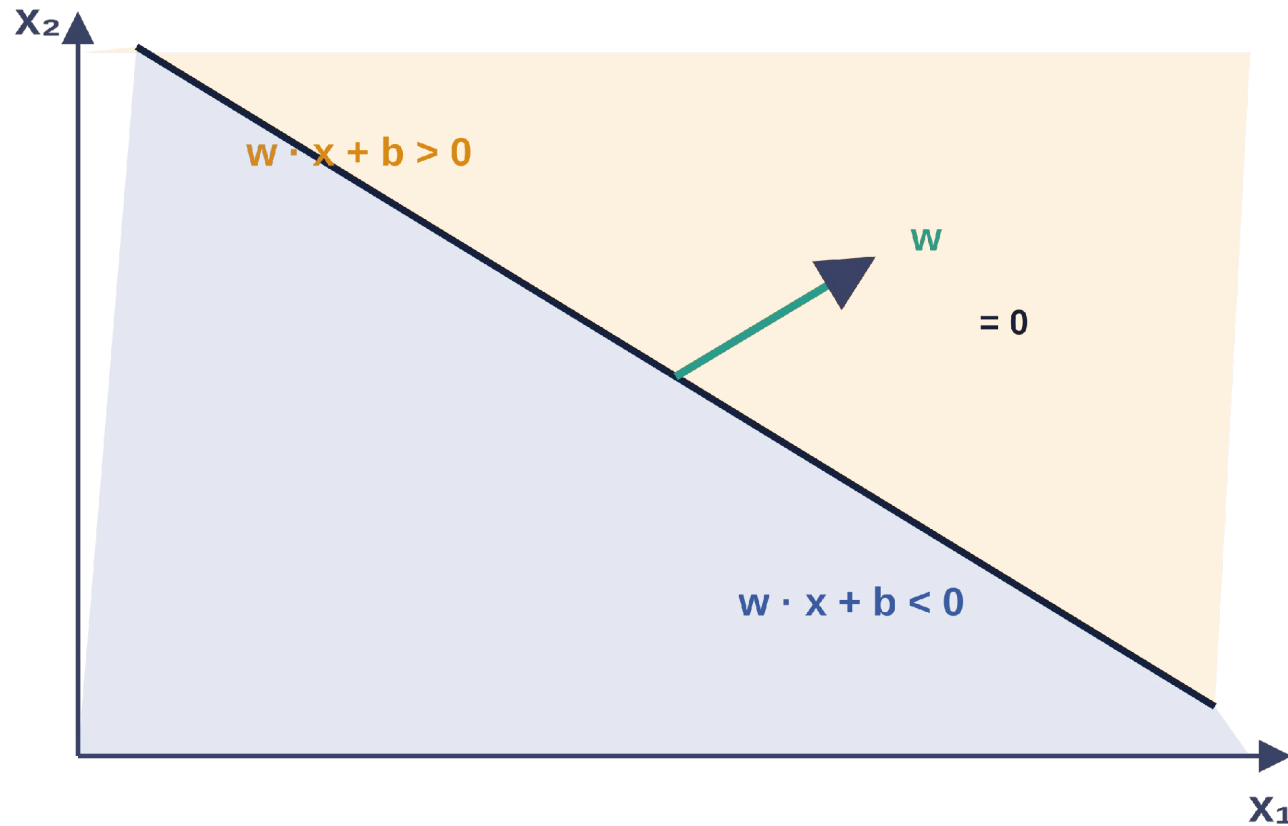


x_1 and x_2 appear only to the first power - **no x^2 , no x_1x_2 .**

There is no curve anywhere in $w \cdot x + b$. The straight line is baked into the formula.

TWO SIDES OF THE LINE

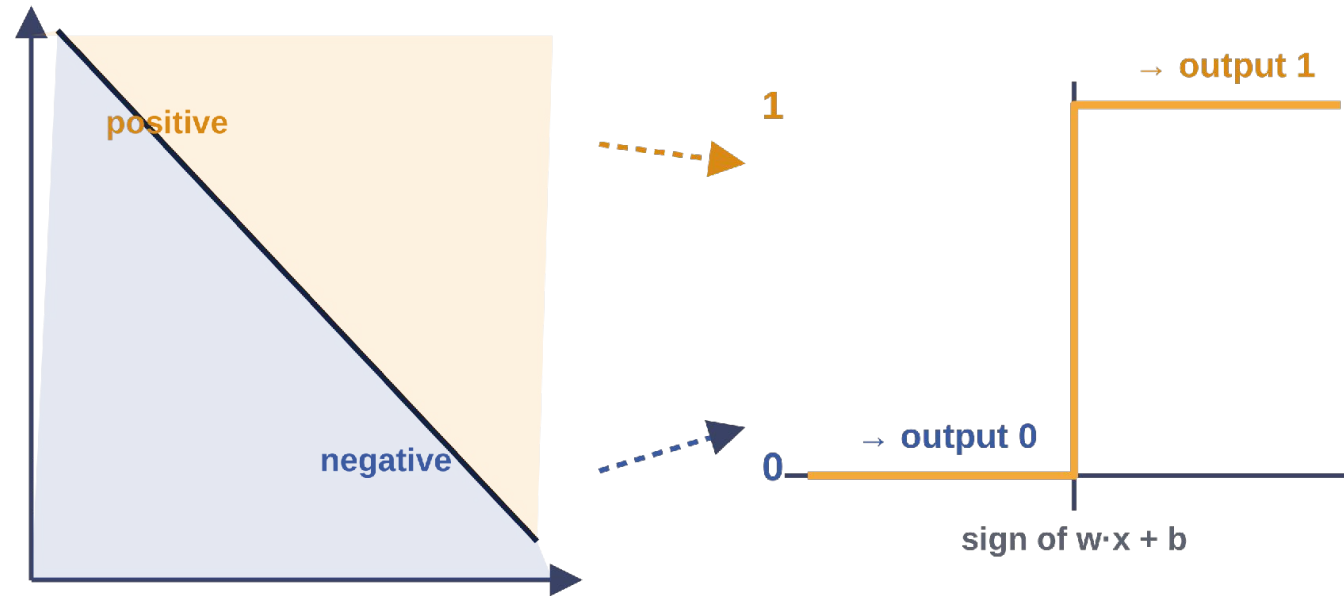
Above, below, and exactly on



- Above the line: the sum is positive
- Below the line: the sum is negative
- On the line: the sum is exactly zero

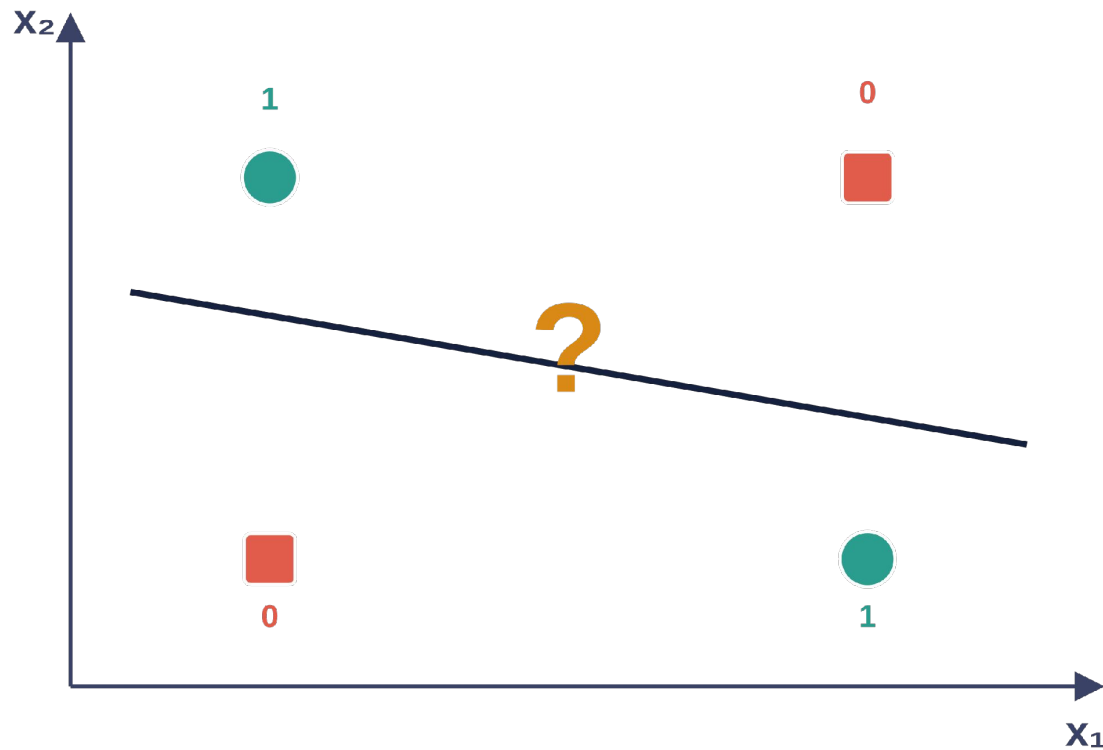
THE PUNCHLINE

The activation only reads the sign



The straight boundary comes entirely from the **linear combination**. **Step or sigmoid, the line stays a line.**

One neuron, one straight line. So what can it never separate?



Meet XOR. No straight line can split it — and in 1969 Minsky & Papert proved it.

Next lecture: we make the weights learn.